

UPDATE

Data mining tools

Andreas Bartschat | Markus Reischl | Ralf Mikut

Institute for Automation and Applied Informatics,
Karlsruhe Institute of Technology, Eggenstein-
Leopoldshafen, Germany

Correspondence

Ralf Mikut, Institute for Automation and Applied
Informatics, Karlsruhe Institute of Technology,
Eggenstein-Leopoldshafen 76344, Germany.
Email: ralf.mikut@kit.edu

This article is an update of: Data mining tools

The development and application of data mining algorithms requires the use of powerful software tools. With challenges such as big data encountered in economy or gene sequencing for life science, data mining is important for daily problems as well as specialized fields. However, the large variety of requirements and user groups lead to a huge number and diversity of software tools. We give an overview by discussing the historical development and presenting a range of existing state-of-the-art data mining and related tools. This paper is an update of our previous article from 2011 following the encyclopedic aspect of Wiley Interdisciplinary Reviews to include new findings or references and changing outdated information. However, since the paper should be able to stand alone, it includes many still valid elements of the previous article. Following the original paper, we propose criteria for the tool categorization based on different user groups, data structures, data mining tasks and methods, visualization and interaction styles, import and export options for data and models, platforms, and license policies. These criteria are then used to classify data mining tools into nine different categories. The typical characteristics of these types are explained and a selection of the most important tools is categorized.

This article is categorized under:

Application Areas > Data Mining Software Tools

KEYWORDS

big data, data analytics, data mining, software tools

1 | INTRODUCTION

Over the last decades, data have become an important part of everyday life. With developments such as big data, they are affecting almost any business area from supply chain and production optimization to marketing and consumer interaction. Smart products become part of our society, generating new benefits and opportunities. Referring to the Cisco forecast, the global IP traffic is expected to reach more than 190 exabyte per month in 2020 (Cisco Visual Networking Index, 2017) and the total amount of data created in 2025 is expected to reach over 160 zettabyte (Reinsel, Gantz, & Rydning, 2017). Given these numbers, the demand and importance of data mining is huge. According to Gartner, in 2016 the worldwide market for business intelligence (BI) accounted over 17 Billion USD and is expected to reach more than 26 Billion USD in 2021, where modern BI and analytics expands more rapidly than other fields (Cheung et al., 2016).

In the past 15–20 years, data mining has become a technology in its own right, is well established, and steadily increases its importance not only in BI but also in technology and life sciences sectors.

For example, in genome sequencing of bacteria (Liu, Hao, Xie, Horsman, & Chen, 2016; Ziemert, Alanjary, & Weber, 2016) or mutation screening in clinical diagnostics research (Pandey, Pabinger, Kriegner, & Weinhäusel, 2017), data mining has become a key factor to characterize compounds as well as to analyze and visualize variants. Social networks and blocks are utilized as a resource for mining sentiments, public opinion tracking or for marketing research (Ravi & Ravi, 2015; Scott, 2017). Further applications can be found in tasks like autonomous driving (Aeberhard et al., 2015) or grasp synthesis for humanoid robots (Bohg, Morales, Asfour, & Kragic, 2014; Levine, Pastor, Krizhevsky, & Quillen, 2017).

As the number of available tools for different areas and tasks continues to grow, the choice of the most suitable tool for a certain use case becomes increasingly difficult for each potential user. This decision-making process can be supported by criteria for the categorization of data mining tools, resulting in different categorizations (Giraud-Carrier & Povel, 2003; Goebel, 1999; Wang, Chen, & Yao, 2008; Wang, Hu, Hollister, & Zhu, 2008). With this work, the authors update their last comprehensive criteria-based survey from 2011 with 269 tools (Mikut & Reischl, 2011). With a focus on recent trends and developments (e.g., big data), the table now covers 498 tools and can be found online.¹ The aim of the paper is not a comprehensive comparison of the listed tools, but a categorization of all tools and the discussion of recent trends. In addition, smaller reviews have been published, for example, covering important open-source tools (Altalhi, Luna, Vallejo, & Ventura, 2017) or analytics in big data and distributed environments (Assunção, Calheiros, Bianchi, Netto, & Buyya, 2015; Grover & Kar, 2017).

Various standardization initiatives have been introduced for data mining processes, data and model interfaces—as with CRISP-DM for industrial data mining (Shearer, 2000), and approaches focused on clinical and biological applications (Mikut, Reischl, Burmeister, & Loose, 2006). A survey of such initiatives is provided by Grossman (Grossman, Hornick, & Meyer, 2002), and a large variety of standard data mining methods is described in comprehensive standard text books (Witten, Frank, Hall, & Pal, 2016); however, new methods, especially for data streams (Bifet, Gavalda, Holmes, & Pfahringer, 2018; Gaber, 2012), extremely large datasets, graph mining (Scott, 2017), text mining (Ravi & Ravi, 2015), and others have been proposed in the last few years.

This paper is organized as follows: the first section highlights the historical development of data mining software until present; the criteria to compare data mining software are explained in the second section. The last section proposes a categorization of data mining software and introduces typical software tools belonging to different categories.

2 | HISTORICAL DEVELOPMENT AND STATE-OF-THE-ART

Data mining has a long history, with strong roots in statistics, artificial intelligence, machine learning, and database research (Fayyad, Piatetsky-Shapiro, & Smyth, 1996; Smyth, 2000). Advances in this field were accompanied by development of related software tools, starting with mainframe programs for statistical analysis in the early 1950's, and leading to a large variety of standalone, client/server, and web-based software as today's service solution. The term “data mining” was coined in the 1980's and gained growing attention from that point on (Lovell, 1983).

Following the original definition, data mining is a step in the knowledge discovery from databases (KDD) process that consists of applying data analysis and discovery algorithms to produce a particular enumeration of patterns (or models) across the data (Fayyad et al., 1996). Furthermore, KDD is defined as the nontrivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data. Sometimes, the wider KDD definition is used synonymously for data mining. This wider interpretation is especially popular in the context of software tools, because most such tools support the complete KDD process, and not just one single step.

With recent developments in processing homogeneous data on distributed systems with batch and stream approaches, a trend away from the term data mining towards data analytics is recognizable. Following the definition of Runkler data analytics describes “the analysis of large data sets for the support of decisions.” Furthermore, it is described as “interdisciplinary field that has adopted aspects from many other scientific disciplines (Runkler, 2016).”

Today, a large number of standard data mining methods is available (Han & Kamber, 2006; Hastie, Tibshirani, & Friedman, 2008). From a historical perspective, these methods have different roots. One early group of methods was adopted from classical statistics: the focus was changed from the proof of known hypotheses to the generation of new hypotheses. Examples include methods from Bayesian decision theory, regression theory, and principal component analysis. Another group of methods stemmed from artificial intelligence like rule-based systems. The term “machine learning” refers to methods that are able to learn from data and includes methods such as decision trees, support vector machines and artificial neural networks. There are several different and sometimes overlapping categorizations; for example, fuzzy logic, artificial neural networks and evolutionary algorithms are summarized as computational intelligence (Engelbrecht, 2007).

The typical life cycle of new data mining methods begins with theoretical papers based on in-house software prototypes, followed by public or on-demand software distribution of successful algorithms as research prototypes. Then, either special commercial or open-source packages containing a family of similar algorithms are developed, or the algorithms are integrated into existing open-source or commercial packages. Many companies have tried to promote their own stand-alone packages, but only few have reached notable market shares. The life cycle of some data mining tools is remarkably short. Typical reasons include internal marketing decisions and acquisitions of specialized companies by larger ones, leading to a renaming and integration of product lines.

The largest commercial success stories resulted from the step-wise integration of data mining methods into established commercial statistical tools. A short historical overview of important players can be found in the author's previous article

(Mikut & Reischl, 2011). In general, statistical tools and databases with integrated analytic capabilities are popular for business application and applied research.

Concurrently, many companies offering BI products have integrated data mining solutions into their database products; one example is Oracle Data Mining (est. 2002). Many of these products are also a product of the acquisition and integration of specialized data mining companies. Equivalents can also be found in the open-source field with databases specialized in machine learning models like MLDB.²

Open-source libraries have also become very popular since the 1990's. The most prominent example is Waikato Environment for Knowledge Analysis (WEKA) (Frank et al., 2005). WEKA started in 1994 as a C++ library, with its first public release in 1996. In 1999, it was completely rebuilt as a JAVA package; since then, it has been regularly updated. In addition, WEKA components have been integrated in many other open-source tools, such as Pentaho, RapidMiner and KNIME.

A large group of research prototypes is based on script-oriented mathematical programs like MATLAB (commercial) and R (open source). Such mathematical programs were not originally focused on data mining, but contain many useful mathematical and visualization functions that support the implementation of data mining algorithms. Recently, graphical user interfaces such as those utilized for R (e.g., Rattle) and MATLAB (e.g., SciXMiner) can be used as integration packages for many single, open-source algorithms.

In 2004, Google presented the programming approach MapReduce, initially used for managing and analyzing the increasing amount of websites. Since this approach was fault-tolerant and scalable to distributed systems, it reached a high popularity. With its integration in the open source project Apache Hadoop, a prototype became available in 2008 and was used by Google and Yahoo! (Dean & Ghemawat, 2008). For offline data processing, it is still used by many companies like Amazon, IBM, and Facebook. Since then, the term big data has been coined, referring not only to the sheer amount of data but also to different data types and fast recording and processing requirements (Wu, Zhu, Wu, & Ding, 2014). This leads to its definition with the three Vs, volume, variety and velocity (Laney, 2001). Extending this definition with the uncertainty of the data and its business value, also a definition with five Vs, including veracity and value, is possible (Ward & Barker, 2013). In 2009, Apache SPARK was initialized to meet these new demands and targeted interactive iterative computations on clusters in contrast to MapReduce for batch processing. By enhancement of the intermediate result storage with in-memory computations and generalization of the MapReduce pattern with a more flexible directed acyclic graph (DAG), SPARK has gained a high popularity (Landset, Khoshgoftaar, Richter, & Hasanin, 2015). Especially the in-memory processing of big data is a key technology for fast and responsive mining and can be found in many commercial products like SAP HANA or SAS products (Gan, Lin, Chao, & Zhan, 2017). For machine learning SPARK offers its own machine learning library MLlib but the open source aspect also allows the development of third party frameworks like H2O. A survey of how big data affects machine learning and classical data mining was published by Japkowicz and Stefanowski (Japkowicz & Stefanowski, 2016).

With recent trends like internet of things (IoT) and smart products or even smart cities, data stream analysis and big data will become an even more relevant topic in the future. Therefore, there will be a great demand for tools allowing to gain value from this data (Rodríguez-Mazahua et al., 2016). Especially the analysis of data streams comes with new challenges such as non-stationary characteristics of the data and dynamic environments including new tools such as MOA (Bifet et al., 2018; Gama, 2010). A survey on data stream analysis with a focus on ensembles can be found in the article of Krawczyk, Minku, Gama, Stefanowski, & Woźniak (2017).

With the increasing processing power of distributed systems, also research in explorative search for data processing pipelines is possible. Summarized by the term AutoML (Automatic Machine Learning), there are first attempts to fully automate the process of algorithm selection and parameterization, like the tool AutoWEKA (Kotthoff, Thornton, Hoos, Hutter, & Leyton-Brown, 2016). Furthermore, the automatic configuration of single methods with high complexity like convolutional neural networks is a topic of research (Feurer et al., 2015; Suganuma, Shirakawa, & Nagao, 2017; Wever, van Rooijen, & Hamann, 2017).

The increasing amount of available data affects not only the tools but also gives rise to analytic methods like deep learning (DL) (LeCun, Bengio, & Hinton, 2015). Especially in benchmarks for image mining these changes became noticeable. In 2012, the first team to use DL was the only team to achieve an error rate of less than 25% in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) (Russakovsky et al., 2015). From this point on, the majority of participants focused on DL strategies and achieved an error rate of less than 5% in 2017. These developments can also be observed in medical image analysis, where contributions of DL have been applied with success and are expected to play an even bigger role in the future (Litjens et al., 2017).

As DL requires not only large datasets for training, but is also computationally expensive, frameworks using graphics processing units (GPU) became popular. The first popular framework in this field was Theano, introducing the concept to express large models as mathematical expressions and using computation graphs for performance and memory optimization (Theano Development Team, 2016). Today, most of the software use a Cuda/C++ based backend and come with APIs for

programming languages like Python, R or MATLAB. Popular frameworks are TensorFlow, initially published in 2015, Caffe (2014) or CNTK (2016) (Abadi et al., 2016; Jia et al., 2014; Seide & Agarwal, 2016). To enhance the implementation speed high level tools like Keras can be used to design new network architectures and for training and evaluation even interactive tools like DIGITS (NVIDIA DIGITS—Interactive Deep Learning GPU Training System) exist.³

Another field affected by big data is graph mining or graph analytics. The analysis of social networks for sentiments and opinions as well as current trends and topics is an important and active field of research. Specialized tools like the standalone software Gephi as well as specialized packages and extensions are available, for example, NetMiner (Python), Statnet (R) or NodeXL (Excel). Especially for recommendations and advertising, there is a requirement for fast methods up to real-time analysis frameworks. Compared to the field of image analysis, a lack of public datasets and benchmarks is noticeable (Sapountzi & Psannis, 2016). Further applications for graph mining can be found in specialized scientific fields like chemical structure analysis. The web tool ChemMine, for example, is specialized for small molecule mining including connectivity to the public database PubChem for chemical compounds (Girke, Cheng, & Raikhel, 2005).

Given the history of data mining tools and their importance in light of current challenges, the number of tools as well as their user group is huge and therefore a categorization is necessary. An attempt for such a categorization will be introduced in the following section.

3 | CRITERIA FOR COMPARING DATA MINING SOFTWARE

In the following, different criteria for comparison of data mining software are introduced. These criteria are based on user groups, the structure of the data, data mining tasks and methods, import and export options, and license models. A detailed overview about the different tools is given later in this paper and as an Excel table in the additional material. However, some specific information is discussed if a specific tool is unique to some aspects of the proposed criteria. The complete list of tools is provided at the end of this paper.

3.1 | User groups

There are many different data mining tools available, which fit the needs of quite different user groups:

- **Business Applications:** This group uses data mining as a tool for solving commercially relevant business applications, such as process optimization and automation, customer relationship management, fraud detection, etc. The increasing individualization of products and integration of consumer and partner requires intelligent and flexible workflows (often summarized with the keyword “Industry 4.0”) (Riel & Flatscher, 2017). This field is mainly covered by a variety of commercial tools providing support for databases with large datasets, and deep integration in the company's workflow.
- **Applied Research:** A user group that applies data mining to research problems, for example technology and life sciences. Here, users are mainly interested in tools with well-proven methods, a graphical user interface (GUI), and interfaces to domain-related data formats or databases.
- **Algorithm Development:** Such users develop new data mining algorithms, and require tools to both integrate their own methods, as well as compare these to existing methods. The necessary tools should contain many competitive algorithms.
- **Education:** For education at universities, data mining tools should be very intuitive, with a comfortable interactive user interface, and inexpensive. In addition, they should allow the integration of in-house methods during programming seminars.

3.2 | Structure of the data

An important criterion is the representation of the underlying potentially high dimensional raw data for the data mining process (Table 1). Based on the nature of the data and the goal of the data mining process the representation of raw data is tailored to the specific task. Furthermore, the same raw data might be presented differently depending on the requirements of different tasks. The first data mining applications were focused on handling datasets represented as two-dimensional feature tables, which is still a common representation in most tools. In this classical format, a dataset consists of a set of N examples (e.g., clients of an insurance company) with s features containing real values or usually integer-coded classes or symbols (e.g., income, age, number of contracts etc.). This format is supported by nearly all existing tools. In some cases, the dataset can be sparse, with only a few non-zero features, such as a list of s shopping items for N different customers. The computational and memory effort can be reduced if a tool exploits this sparse structure.

TABLE 1 Maximum dimensions of datasets for different types of problems

Data	Dim.	Structure for each of the N examples
Feature table	2	s features (e.g., age, income)
Texts	2	Frequency of words or n -grams (vector-space approach)
Event logs	2	K event information associated with a time stamp
Time series	3	s time series with K time samples
Sequences	3	s sequences of length L (e.g., mass spectrograms, genes)
Audio	3	s sampled sound waves with K time samples
Images	4	s color images with pixels
Graphs	4	s graphs with adjacency matrices
3D images	5	s images with pixels and slices
Videos	5	s videos containing images with pixels and K time samples
3D videos	6	Like videos, but with additional slices

Notes. Lower dimensions of the dataset can occur for problems with only one feature $s = 1$ resp. one example ($N = 1$). Dim., maximum dimensions of a dataset, including s number of features; N , number of examples; K , number of samples in a time series.

Some structured datasets are characterized by the same dimensionality. As an example, sample documents in most text mining problems are represented by the frequency of words or so-called n -grams (a group of n subsequent characters in a document) (Damashek, 1995). Further special cases of low dimensional data are digital representations of audio signals with K sampling points and event logs, representing event information occurring not necessarily uniformly over time and therefore need an association with time stamps.

The most prominent format having a higher dimensionality contains time series as elements leading to dataset dimensionality between one (i.e., only one example of a time series with K samples) and three (i.e., N different examples of s -dimensional vector time series with K samples). Typical tasks are forecasting of future values, finding typical patterns in a time series or finding similar time series by clustering. The analysis of time series plays an import role in many different applications, including prediction of stock markets, forecasting of energy consumption and other markets, and quality supervision in production. It is also supported by most data mining tools.

With a similar dimensionality, different kinds of structured data exist, such as gene sequences (spatial structure), spectrograms or mass spectrograms (structured by frequencies or masses). Only a few tools support these types of structured data explicitly, but some tools for time series analysis can be rearranged to cope with these problems.

A more recent trend is the application of data mining methods for images and videos with goals like image tagging and labeling as well as object detection and semantic segmentation (Kim et al., 2017; Liu, Chen, & Kubota, 2013). The main challenge is the handling of extremely large raw datasets, up to giga- and terabytes, caused by the high volume of examples. Typical applications are microscopic images in biology and medicine, camera-based sensors in quality control, autonomous driving, robotics, biometrics and security. Such datasets must supported by metadata—with links to image and video files handled in a main dataset, and files—which contain the main part of the data. Until now, these problems were usually solved using a combination of tools: the initial tool (e.g., ImageJ, ITK) would process the images or videos, resulting in segmented images and extracted features describing the segments; a second tool would solve data mining problems handling the extracted features as a classical table or time series (Bartschat, Hübner, Reischl, Mikut, & Stegmaier, 2016; Gaber, 2012; Schindelin, Rueden, Hiner, & Eliceiri, 2015).

Another format leading to image-like dimensions includes graphs that can be represented as adjacency matrices, describing the connection between different nodes of a graph. Graph mining has powerful applications (Chakrabarti & Faloutsos, 2006; Borgelt, 2009), such as characterizing social networks and chemical structures, with various tools such as Cytoscape, Centrifuge or Gephi. In general, modern data mining tools tend to handle and combine a variety of different data types.

Other than that, also semi structured data like JSON or XML become more popular and are supported by an increasing number of tools.

3.3 | Tasks and methods

The creation of models based on the available data are machine learning tasks such as forecasting or trend analysis, clustering, categorization, structure discovery, novelty detection and labeling or tagging of examples as well as segmentation. The most important tasks in data mining are:

- supervised learning, with a known output variable in the dataset, including
 - classification: class prediction, with the variable typically coded as an integer output,

- probabilistic or fuzzy classification: gradual values between 0 and 1 applied to the different classes, either in a probabilistic (probability of the given class) or possibilistic interpretation (gradual membership to different classes),
- regression: prediction of a real-valued output variable, including special cases of predicting future values in a time series out of recent or past values;
- unsupervised learning, without a known output variable in the dataset, including
 - clustering: finds and describes groups of similar examples in the data using crisp or fuzzy clustering algorithms,
 - association learning: finds typical groups of items that occur frequently together in examples;
- semi-supervised learning, whereby the output variable is known only for some examples;
- reinforcement learning, whereby the output variable is unknown and only a reward to an action is known (Kober, Bagnell, & Peters, 2013).
- active learning, whereby a new example including its output variable can be acquired by the algorithm on demand for a limited amount of examples.

Each of these tasks consists of a customized pipeline with different modules. Examples of such operations are:

- data cleaning (e.g., outlier detection),
- data filtering (e.g., smoothing of time series),
- feature extraction from time series, images, videos, and graphs (e.g., consisting of segmentation and segment description for images, characteristic values such as community structures in graphs),
- feature transformation (e.g., mathematical operations, including discretization, logarithms, dimension reduction by linear or nonlinear combinations, factor analysis or independent component analysis),
- feature evaluation and selection (e.g., filter or wrapper methods),
- computation of similarities and detection of the most similar elements in terms of examples or features (e.g., *k*-nearest-neighbor-methods, correlation analysis),
- model validation (cross validation, bootstrapping, statistical relevance tests, complexity measures),
- model fusion (mixture of experts), and
- model optimization (e.g., evolutionary algorithms).

With advances in DL, the integration of various modules (e.g., data filtering, feature extraction) into an integrated module becomes a promising option. Alternatively, modules can be automatically selected and parameterized by meta-learning approaches or evolutionary algorithms (LeCun et al., 2015; Suganuma et al., 2017). For almost all of these tasks, a large variety of classical statistical methods—including classifiers using estimated probability density functions, factor analysis and others, and machine learning methods—such as artificial neural networks, fuzzy models, rough sets, support vector machines, decision trees, and random forests are available. In addition, optimization models such as evolutionary algorithms can assist with the identification of model structures and parameters. The related methods are described in survey articles or textbooks and are not summarized in this paper (Han & Kamber, 2006; Hastie et al., 2008; Jain, Duin, & Mao, 2000).

Not all of the data mining methods are available in all software tools. The following list contains a subjective evaluation of the frequency with which specific methods are incorporated in the different tools:

- Common: data cleaning, data filtering, feature extraction, principal component analysis, factor analysis, advanced feature evaluation and selection, statistical feature selection and relevance tests, *k*-nearest-neighbor-methods, correlation analysis, clustering, regression, classifiers using estimated probability density functions, decision trees, random forests, artificial neural networks, model cross validation, support vector machines,
- Sometimes: association learning and mining frequent item sets, independent component analysis, complexity measures, fuzzy classification, Bayesian networks, learning of crisp rules (if/then rules that are applied if the rule premise is true), model fusion, bootstrapping,
- Rare: learning of fuzzy systems (contained in KnowledgeMiner, See5, SciXMiner), and model optimization by evolutionary algorithms (in ECJ, KEEL, D2K) (Alcalá-Fdez et al., 2009).

3.4 | Interaction and visualization

There are four main types of interaction between a user and a data mining tool:

- pure textual interface using a programming language—difficult to handle, but easily automated,
- graphical interface with a menu structure—easy to handle, but not so easily automated,

- graphical user interface where the user selects “function blocks” or algorithms from a palette of choices, defines parameters, places them in a work area, and connects them to create complete data mining streams or workflows; a good compromise, but difficult to handle for large workflows, and
- visualization interface like dashboards with a collection of different charts and overlays to display various data types and results.

Mixtures of these forms arise if macros of menu items can be recorded for workflows, or if additional blocks in a workflow can be implemented using a programming language. Automation (scripting) is extremely important for routine tasks, especially with large datasets, because the workload of the user is reduced. Almost all tools provide powerful visualization techniques for the presentation of data mining results; particularly tools for business application and applied research that are able to generate complete reports containing the most important results in a readable form for users lacking explicit data mining skills. Interactive methods can support an explorative data analysis. More details on visualization and interactive knowledge discovery can be found in books and review articles (Simoff, Böhlen, & Mazeika, 2008; Stahl, Gabrys, Gaber, & Berendsen, 2013; Sun, Wu, Liang, & Liu, 2013). For some fields, even specialized reviews exist, for example, for social media data (Wu, Cao, Gotz, Tan, & Keim, 2016).

3.5 | Import and export of data and models

The ease with which data and models can be imported and exported among different software tools plays a crucial role in the functionality of data mining tools. First, data is usually generated and hosted from different sources, such as databases or software associated with measurement devices. With an increasing amount, complexity and variety of data, the importance of unified standards for data storage and access grows as, for example, copying or transforming data is not practicable. Especially for business applications, a narrow coupling of “not only structured query language” (NoSQL) databases with commercial tools as well as open-source solutions (Apache Ecosystem) is beneficial. Since almost all other non-data mining tools support export as text or Excel files, formats like comma separated values (CSV), JavaScript object notation (JSON) or extensible markup language (XML) are used import formats in data mining tools. In addition, almost all software has proprietary binary or textual files, and exchanges formats for data and models, for example, ARFF (WEKA standard).

In order to import and export developed models as components in other processes and systems, the XML-based standard PMML was initiated by the Data Mining Group,⁴ (Pechter, 2009) and continuously developed with its latest release of version 4.3 in 2016. It is supported by many companies like IBM and SAS as well as tools like KNIME and Apache SPARK. As a complementary for JSON, the Data Mining Group proposed the standard PFA. Another standard initiative is the Object Linking and Embedding Database (OLE DB, sometimes written as OLEDB or OLE-DB) for data mining, an Application Programming Interface (API) designed by Microsoft to access different types of data stored in a uniform manner.⁵ OLE DB is a set of interfaces implemented using the Component Object Model (COM). For data exchange among different tools, another initiative deals with Java Specification Requests for data mining: Versions 1.0 (JSR 73, final release in 2004)⁶ defines an extensible Java API for data mining systems. The consortium includes many related companies, such as Oracle, SAS, IBM, SAP and others (Anand et al., 2007; Hornick, Marcadé, & Venkayala, 2006). Another interesting feature is the export of an executable runtime version of developed models. Often, they do not require a more expensive development license and can be run free of charge, or at least with a cheaper runtime license. However, with an increasing role of databases and proprietary formats for fast caching and distribution in cluster systems tailored for in-memory processes, also proprietary standards and workflows arise.

3.6 | Platforms

Data mining tools can be subdivided into stand-alone and client/server solutions. Client/server solutions dominate, especially in products designed for business users. They are available for different platforms, including Windows, MAC OS, Linux and special mainframe supercomputers. There is a growing number of JAVA-based systems that are platform-independent for users in research and applied research.

Further trends are cloud-based SAAS (software as a service) products like IBM Watson or the Google Cloud Platform. However, this trend has the potential risk of hurting privacy policies, since the protection of data is difficult and many companies are very careful with sensitive data.

3.7 | Licenses

There is a wide variety of commercial and open-source licenses for data mining tools. This is particularly true in the business application user group, where commercial software is very attractive due to high software stability, good coupling with other

commercial tools for data warehouses, included software maintenance, and the possibility of user training for sophisticated topics. For all other user groups, there is a strong trend towards open-source software, but different types of licenses exist (Sonnenburg et al., 2007). The main advantages of open-source software are faster bug fixes and methodological improvements, potential for integration with other tools, the existence of developer and user communities, faster adoption of methods to other innovative applications, and the fair comparison of new data mining algorithms with alternative ones. These advantages attract mainly users of applied research, development and education; however, open-source tools are beginning to migrate even into business user groups, particularly if additional commercial services, such as training or maintenance are offered (e.g., Pentaho) (Bitterer, 2009).

The most popular open-source licenses are the MIT License, the Apache License 2.0 and the GNU GPL v3.⁷ One major aspect of licenses are the restrictions when it comes to usage, modification and redistribution. The GNU GPL is a representative of copyleft licenses, which permits free redistribution, integration in other packages, and modification of the software as long as all subsequent users receive the same freedoms. Meaning, it can only be used in open-source projects. The other two licenses are permissive and thus less restrictive, allowing their integration in commercial and closed-source products.⁸

Not only the free availability of algorithms but also of data sets is critical for reproducible research. Initiatives such as OpenML⁹ and platforms like Kaggle¹⁰ for comparison of algorithms and data sets are important developments in this regard. Furthermore, tools like KNIME allow the publication and exchange of whole processing pipelines supporting the idea of open science (Fillbrunn et al., 2017; Vanschoren, Braun, & Ong, 2014).

4 | CATEGORIZATION OF DATA MINING SOFTWARE INTO DIFFERENT TYPES

Following the criteria from the previous section, different types of similar data mining tools can be found. The typical characteristics of these types are explained in this section. Matching of the different types and user groups and the number of recent tools are summarized in Table 2. In addition, for commercial data mining tools, related tools and their group membership is summarized in different tables for commercial (Table 3) and free and open-source data mining tools (Table 4). In these tables, very popular tools are marked in bold. The popularity was measured by:

- the 24 products listed in the Magic Quadrant 2017 from Gartner (Sallam et al., 2017),
- the top tools of the “New Leader, Trends, and Surprises in Analytics, Data Science, Machine Learning Software Poll” (2017) of KDnuggets with 2,900 voters,¹¹
- the 24 most frequently used tools by data scientists listed by the “Data Science Survey 2015” from Rexer (Rexer, Gearan, & Allen, 2016),
- the 10 most frequently used tools as well as tools with a high increase from “Analytics, Data Mining software” (2015) of KDnuggets with nearly 3,000 voters,¹² and
- the most popular image processing tools (ITK and ImageJ) from the author's own experience to cover this field.

In this paper, we use a slightly modified version of our previous categorization (Mikut & Reischl, 2011), the following nine types are proposed:

- *Data mining suites (DMS)* focus largely on data mining and include numerous methods. They support feature tables and time series, while additional tools for text mining are sometimes available. The application focus is wide and not restricted to a special application field, such as business applications. However, coupling to business solutions, import and export of models, reporting, and a variety of different platforms are supported. In addition, the producers provide services for adaptation of the tools to the workflows and data structures of the customer. DAS are mostly commercial and rather expensive with typical examples like IBM SPSS Modeler, SAS Enterprise Miner, SAP Predictive Analytics, Knowledge Studio,

TABLE 2 Matching between different user groups and tool types with number of recent tools in the Excel table (see Supplementary Information section, tools belonging to two categories are counted twice)

Types	DMS	BI	MAT	INT	EXT	LIB	SPEC	RES	SOL
Number of recent tools	72	66	17	10	10	37	89	5	35
Business Applications	+	+	–	0	0	–	0	–	0
Applied Research	+	–	+	+	0	0	0	0	+
Algorithm Development	–	–	+	+	–	+	0	+	–
Education	+	–	0	0	+	–	+	–	0

Note. Evaluation: +, especially useful; 0, less useful; –, Not useful.

TABLE 3 Commercial data mining tools

Tool	Type	Link
Alteryx	DAS	www.alteryx.com
Analytics OS	DAS, BI	www.pyramidanalytics.com
Birst	BI	www.birst.com
Board	BI	www.board.com
ClearStory Data	BI	www.clearstorydata.com
Datameer	BI	www.datameer.com
Domo	BI	www.domo.com
Einstein Analytics Platform	BI	www.salesforce.com
IBM Cognos Analytics	BI	www.ibm.com/analytics/products/
IBM SPSS	DAS	www.ibm.com/analytics/products
IBM Watson	DAS	www.ibm.com/watson-analytics
Logi Analytics Platform	DAS	www.logianalytics.com
MATLAB	MAT	www.mathworks.com
Mathematica	MAT	www.wolfram.com/mathematica
Microsoft Power BI	BI	https://powerbi.microsoft.com
MicroStrategy	BI	www.microstrategy.com
Oracle Visualization and Big Data Products	DAS	www.oracle.com/big-data/products.html
Pentaho	BI	www.sourceforge.net/projects/pentaho
Qlik Analytics Platform	BI	www.qlik.com/us/products/qlik-analytics-platform
SAP BusinessObjects	BI	www.sap.com/products/analytics/business-intelligence-bi.html
SAS Visual Analytics	DAS	www.sas.com/en_us/software/visual-analytics.html
Sisense	BI	www.sisense.com
Tableau	BI	www.tableausoftware.com
TIBCO Spotfire	DAS	spotfire.tibco.com
Teradata Aster Analytics	BI	www.teradata.com/products-and-services/teradata-analytics-platform
ToughSpot	DAS	www.thoughtspot.com
WebFOCUS	BI	www.informationbuilders.com
Yellowfin BI	BI	www.yellowfinbi.com
Zoomdata	DAS	www.zoomdata.com

Note. Very popular tools are marked in bold letters.

TABLE 4 List of free and open-source data mining tools

Tool	Type	Link
Apache Spark	MAT	https://spark.apache.org
Apache Flink	LIB	https://flink.apache.org
H2O	SPEC	www.h2o.ai
ImageJ	SOL	https://rsbweb.nih.gov/ij
ITK	SOL	www.itk.org
KEEL	INT	www.keel.es
KNIME	DAS	www.knime.org
Orange	DAS	https://orange.biolab.si
OpenCV	LIB	www.opencv.org
Pegasus	RES	www.cs.cmu.edu/~pegasus
Pentaho	BI	www.pentaho.com
Python	MAT	www.python.org
R	MAT	www.r-project.org
RapidMiner	DAS	www.rapidminer.com
Scikit-learn	LIB	www.scikit-learn.org
TensorFlow	SPEC	www.tensorflow.org
WEKA	DAS, LIB	www.cs.waikato.ac.nz/ml/weka

Note. Very popular tools are marked in bold letters.

NAG Data Mining Components, Partek Discovery Suite, STATISTICA and TIBCO Spotfire. Furthermore, also some open-source tools such as KNIME and RapidMiner exist.

- *BI packages* have no special focus to data mining, but include basic data mining functionality, especially for statistical methods, for example, forecasting and trend lines by regression models or clustering in business applications. They have a highly developed reporting and visualization functionality like dashboards and good support for education, handling, and adaptation to the workflows of the customer. They are characterized by a strong focus on database coupling or they consist of databases with integrated analytic and prediction features, mostly following a client/server architecture. Most BI software is commercial (Tableau, IBM Cognos, SAP BusinessObjects, Teradata Database), but a few open-source solutions exist (SpagoBI, Pentaho).
- *Mathematical (MAT) packages* have no special focus on data mining, but provide a large and extendable set of mathematical algorithms and visualization routines. They support feature tables, time series and have at least import formats for images. The user interaction often requires programming skills in a scripting language. MATs are attractive to users in algorithm development and applied research because data mining algorithms can be rapidly implemented, mostly in the form of extensions (EXT) and research prototypes (RES). This definition includes programming languages which provide a similar user experience as tools with small scripts and usage of the provided libraries. This definition includes programming languages which can be used as a tool with small scripts and usage of the provided libraries. A popular commercial representative is MATLAB but also the popularity of open-source tools like R and Python (with a large number of data mining packages) are steadily increasing (Rexer et al., 2016). Most tools are available for different platforms but have weaknesses in database coupling.¹³
- *Integration (INT) packages* are extendable bundles of many different open-source algorithms, either as stand-alone software (mostly based on Java; as the GUI-version of WEKA) or as a kind of larger extension package for tools from the MAT type (like scikit-learn for Python, SciXMiner for MATLAB, and RWEKA for R). Import and export support standard formats, but database support is weak. Most tools are available for different platforms and include a GUI. Mixtures of license models occur if open-source integration packages are based on commercial tools from the MAT type. With these characteristics, the tools are attractive to algorithm developers and users in applied research due to expandability and rapid comparison with alternative tools, and due to simple integration of application-specific methods and import options.
- *EXT* are smaller add-ons for other tools like Excel, MATLAB, R, etc. with limited, but useful functionality. Here, only a few data mining algorithms are implemented, such as artificial neural networks for Excel (Analytic Solver) or MATLAB (MATLAB Neural Networks Toolbox). There are commercial or open-source versions, but licenses for the basic tools must also be available. The user interaction is the same as for the basic tool, for example by using a programming language (MATLAB) or by embedding the extension in the menu (Excel).
- *Data mining libraries (LIB)* implement data mining methods as a bundle of functions. These functions can be embedded in other software tools using an Application Programming Interface (API) for the interaction between the software tool and the data mining functions. A GUI is missing, but some functions can support the integration of specific visualization tools. Most commonly, they are programmed in JAVA or C++ and usually come with interfaces to other languages like Python or R. Open source examples are WEKA (Java-based), MLC++ (C++ based), and LibSVM (C++ based) for support vector machines. A commercial example is Neurofusion for C++, while XELOPES (Java, C++, C#) uses different license models. LIB tools are mainly attractive to users in algorithm development and applied research, for embedding data mining software into larger data mining software tools or specific solutions for narrow applications.
- *Specialties (SPEC)* are similar to DAS tools, but implement only one special family of methods, such as artificial neural networks. They contain many elaborate visualization techniques for such methods. SPECs are rather simple to handle compared to other tools, which eases the use of such tools in education. Examples are CART for decision trees, Bayesia-Lab for Bayesian networks, C5.0, Rule Discovery System for rule-based systems, and TensorFlow and Neuroshell for neural networks.
- *RES* are usually the first—and not always stable—implementations of new and innovative algorithms. They contain only one or a few algorithms with restricted graphical support and without automation support. Import and export functionality is rather restricted and database coupling is missing or weak. RES tools are mostly open-source. They are mainly attractive to users in algorithm development and applied research, specifically in very innovative fields. Examples are GIFT for content-based image retrieval. Early versions of today's popular tools like WEKA and RapidMiner started in this category and shifted later to other categories as DAS.
- *Solutions (SOL)* describe a group of tools that are customized to narrow application fields such as text mining (GATE), image processing (ITK, ImageJ), image analysis in microscopy (CellProfilerAnalyst), or mining gene expression profiles (Partek Genomics Suite, MEGA). The advantage of these solutions is the excellent support of domain-specific feature extraction techniques, evaluation measures, visualizations, and import formats. The level of data mining methods ranges

from rather weak support (particularly in image processing), up to highly developed algorithms. In some cases, more general tools from types DAS or INT also support specific domains (KNIME, SciXMiner for peptide chemoinformatics (Mikut, 2010)). There are many commercial and open-source solutions.

A large variety of tools actually requires a fuzzy categorization with gradual memberships to different types. Examples are tools including a set of different algorithms (LIB) with an additional GUI acting as an INT package, DMS with many integrated open source algorithms including a GUI (similar to INT), but with many own algorithms (KNIME), including special methods for narrow application fields and others. In these cases, a main type was assigned and the other fuzzy memberships are discussed in the Excel table in the Supporting Information section.

The following kinds of tools were not included in the comparison:

- non-available software (e.g., due to company mergers, stopped developments) is only listed in the Excel table in the Supporting Information section,
- software for the handling of data warehouses without explicit focus on data mining,
- software for the manual design and application of rule-based systems,
- software for table calculation with a focus to office users,
- customized solutions for very narrow fields, and
- internal software solutions of data mining consulting companies (software as a service provider).

5 | CONCLUSIONS

Many advanced tools for data mining are available either as open-source or commercial software. They cover a wide range of software products, from problem-independent DMS and business-centered data warehouses with integrated data mining capabilities to early research prototypes for newly developed methods. In this paper, nine different types of tools are presented: data mining software, BI packages, mathematical packages, INT packages, extensions, specialties, research prototypes, data mining libraries, and solutions. They vary in many different characteristics, such as intended user groups, possible data structures, implemented tasks and methods, interaction styles, import and export capabilities, platforms, and license policies.

Because the selection of the most relevant tool strongly depends on the user's needs, a decision support is very difficult. We recommend a selection of the most appropriate tool category and then a systematic comparison of popular tools within this category.¹⁴ A reasonable starting point is the search for similar use cases in whitepapers and scientific articles. If the user wishes to cover a large set of different use cases, MAT packages with a large user community as R, Python or MATLAB are recommended.

New trends in data mining lead to the development of new tools or prototypes. A recent example is deep neural networks that lead to the rise of various new tools like TensorFlow or distributed computation with tools like SPARK. Only a few tools manage to become a standalone tool and successful algorithms will be integrated in existing tools. The last point is especially true for more general algorithms such as the support vector machine for classification, which today can be found in almost every data mining suite. For these reasons, the field of data mining tools changes rapidly and maintaining an overview is almost impossible. Thus, it is recommended to focus on established tools for routine tasks and on rising tools to follow research trends.

New generalized tools evolve when highly specialized approaches can be grouped and integrated in a new framework. One active field of research is the analysis of complex data structures. The overview given by Battaglia et al. (2018) covers structured data with a focus on graphs with a broad range of applications, including general problems like social network analysis, but also highly specialized topics like the simulation of quantum mechanics (DFT) (Becke, 1993; Gilmer, Schoenholz, Riley, Vinyals, & Dahl, 2017). Since this is an active research topic, we expect new and the extensions of available tools to cover such data structures.

Another key point is the systematic analysis of heterogeneous data, not integrated in a data warehouse. One example is the concept of data lakes covering many datasets in various formats and methods to retrieve and connect them, for example, by metadata (Hai, Geisler, & Quix, 2016; O'Leary, 2014). This field also requires new methods and is expected to be active in the future. The same is true for the development of distributed machine learning algorithms for high performance computing (HPC) on distributed systems.

Trends like AutoML strive to make routine tasks in data mining easily accessible for non-experts. With a focus on automated model selection, AutoML simplifies the integration of data mining in commercial workflows and applications. However, since the computational complexity for AutoML is still very high, it requires further research. Nevertheless, as these paradigms become more common, we expect their integration in available tools.

Another active field is the deployment of smart devices. Gartner predicts around 20 Billion internet of things (IoT) devices by 2020.¹⁵ As these devices are expected to not only collect data but also process them with data mining techniques and artificial intelligence, new unified IoT frameworks and solutions are required. These decentralized systems bring up new challenges in terms of data processing and other fields, for example, privacy and security. Thus, also the integration of new paradigms like the blockchain will become more and more important for data mining tools (Dorri, Kanhere, Jurdak, & Gauravaram, 2017).

Together with the decentralization and the connection of distributed devices with different operating systems, also standardization and interfaces (APIs) will become more important. Such standardization initiatives will also be competitive advantages for service providers for Big Data and Cloud solutions, by enabling services like data analytics or visualization as a service (Assunção et al., 2015).

In summary, there are many active research fields and new requirements for data mining. Thus, the increase of tools will continue in the future and finding the right tool for a specific use case will remain a challenging task.

CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

ENDNOTES

¹Supplementary Excel Table: <https://sourceforge.net/projects/data-analytics-tools/>.

²Machine Learning Database: <https://mldb.ai>.

³<https://developer.nvidia.com/digits>.

⁴www.dmg.org.

⁵www.msdn.microsoft.com/en-us/library/ms146608.aspx.

⁶www.jcp.org/en/jsr/detail?id=73.

⁷See www.blackducksoftware.com/top-open-source-licenses and <https://github.com/blog/1964-open-source-license-usage-on-github-com>.

⁸www.choosealicense.com.

⁹www.openml.org.

¹⁰www.kaggle.com.

¹¹www.kdnuggets.com/2017/05/poll-analytics-data-science-machine-learning-software-leaders.html.

¹²www.kdnuggets.com/polls/2015/analytics-data-mining-data-science-software-used.html.

¹³In principle, table calculation software like Excel may also be categorized here, but it is not included in this paper.

¹⁴Supplementary Excel Table: <https://sourceforge.net/projects/data-analytics-tools/>.

¹⁵Source: Gartner.

RELATED WIREs ARTICLE

[Software and applications of spatial data mining](#)

REFERENCES

- Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., ... Zheng, X. (2016). TensorFlow: Large-scale machine learning on heterogeneous systems. <https://www.tensorflow.org/>.
- Aeberhard, M., Rauch, S., Bahram, M., Tanzmeister, G., Thomas, J., Pilat, Y., ... Kaempchen, N. (2015). Experience, results and lessons learned from automated driving on Germany's highways. *IEEE Intelligent Transportation Systems Magazine*, 7(1), 42–57.
- Alcalá-Fdez, J., Sánchez, L., García, S., del Jesus, M. J., Ventura, S., Garrell, J. M., ... Herrera, F. (2009). KEEL: A software tool to assess evolutionary algorithms for data mining problems. *Soft Computing*, 13(3), 307–318.
- Altalhi, A. H., Luna, J. M., Vallejo, M., & Ventura, S. (2017). Evaluation and comparison of open source software suites for data mining and knowledge discovery. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 7(3), e1204.
- Anand, S. S., Grobelnik, M., Herrmann, F., Hornick, M., Lingensfelder, C., Rooney, N., & Wettschereck, D. (2007). Knowledge discovery standards. *Artificial Intelligence Review*, 27(1), 21–56.
- Assunção, M. D., Calheiros, R. N., Bianchi, S., Netto, M. A., & Buyya, R. (2015). Big data computing and clouds: Trends and future directions. *Journal of Parallel and Distributed Computing*, 79, 3–15.
- Bartschat, A., Hübner, E., Reischl, M., Mikut, R., & Stegmaier, J. (2016). XPIWIT—An XML pipeline wrapper for the insight toolkit. *Bioinformatics*, 32(2), 315–317.

- Battaglia, P. W., Hamrick, J. B., Bapst, V., Sanchez-Gonzalez, A., Zambaldi, V., Malinowski, M., ... Pascanu, R. (2018). Relational inductive biases, deep learning, and graph networks. *ArXiv preprint arXiv: 1806.01261*.
- Becke, A. D. (1993). Density-functional thermochemistry. III. The role of exact exchange. *The Journal of Chemical Physics*, 98(7), 5648–5652.
- Bifet, A., Gavaldà, R., Holmes, G., & Pfahringer, B. (2018). *Machine learning for data streams with practical examples in MOA*. Cambridge: MIT Press.
- Bitterer, A. (2009). Open-source business intelligence tool production deployments will grow five-fold through 2010. *Gartner RAS Research Note*, G00171189.
- Bohg, J., Morales, A., Asfour, T., & Kragic, D. (2014). Data-driven grasp synthesis—a survey. *IEEE Transactions on Robotics*, 30(2), 289–309.
- Borgelt, C. (2009). Graph mining: An overview. In *Proceedings 19. GMA/GI Workshop Computational Intelligence* (pp. 189–203). Karlsruhe, Germany: KIT Scientific Publishing.
- Chakrabarti, D., & Faloutsos, C. (2006). Graph mining: Laws, generators, and algorithms. *ACM Computing Surveys (CSUR)*, 38(1), 1–69.
- Cheung, M., Dharmasthira, Y., Eschinger, C., Anderson, R. P., Granetto, B. F., Adams, A., ... Roth, C. (2016). *Forecast: Enterprise software markets, worldwide, 2013–2020*. Stamford, CT: Gartner RAS Core Research Notes Gartner.
- Cisco Visual Networking Index. (2017). *The zettabyte era: trends and analysis*. San Jose, CA: Cisco.
- Damashek, M. (1995). Gauging similarity with n-grams: Language-independent categorization of text. *Science*, 267(5199), 843–848.
- Dean, J., & Ghemawat, S. (2008). MapReduce: Simplified data processing on large clusters. *Communications of the ACM*, 51(1), 107–113.
- Dorri, A., Kanhere, S. S., Jurdak, R., & Gauravaram, P. (2017). Blockchain for IoT security and privacy: The case study of a smart home. In *2017 IEEE international conference on pervasive computing and communications workshops (PerCom Workshops)* (pp. 618–623). Piscataway, NJ: IEEE.
- Engelbrecht, A. P. (2007). *Computational intelligence—An introduction*. Chichester, England: Wiley.
- Fayyad, U., Piatetsky-Shapiro, G., & Smyth, P. (1996). From data mining to knowledge discovery in databases. *AI Magazine*, 17, 37–54.
- Feurer, M., Klein, A., Eggensperger, K., Springenberg, J., Blum, M., & Hutter, F. (2015). Efficient and robust automated machine learning. In *Advances in neural information processing systems* (pp. 2962–2970). Red Hook, NY: Curran Associates, Inc.
- Fillbrunn, A., Dietz, C., Pfeuffer, J., Rahn, R., Landrum, G. A., & Berthold, M. R. (2017). KNIME for reproducible cross-domain analysis of life science data. *Journal of Biotechnology*, 261, 149–156.
- Frank, E., Hall, M. A., Holmes, G., Kirkby, R., Pfahringer, B., & Witten, I. H. (2005). Weka: A machine learning workbench for data mining. In *Data mining and knowledge discovery handbook: A complete guide for practitioners and researchers* (pp. 1305–1314). New York: Springer.
- Gaber, M. M. (2012). Advances in data stream mining. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 2(1), 79–85.
- Gama, J. (2010). *Knowledge discovery from data streams*. New York: Chapman and Hall/CRC.
- Gan, W., Lin, J. C.-W., Chao, H.-C., & Zhan, J. (2017). Data mining in distributed environment: A survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 7(6), e1216.
- Gilmer, J., Schoenholz, S. S., Riley, P. F., Vinyals, O., & Dahl, G. E. (2017). Neural message passing for quantum chemistry. In *International conference on machine learning* (pp. 1263–1272). PMLR.
- Giraud-Carrier, C., & Povel, O. (2003). Characterising data mining software. *Intelligent Data Analysis*, 7(3), 181–192.
- Girke, T., Cheng, L.-C., & Raikhel, N. (2005). ChemMine. A compound mining database for chemical genomics. *Plant Physiology*, 138(2), 573–577.
- Goebel, M. (1999). A survey of data mining and knowledge discovery software tools. *ACM SIGKDD Explorations Newsletter*, 1(1), 20–33.
- Grossman, R. L., Hornick, M. F., & Meyer, G. (2002). Data mining standards initiatives. *Communications of the ACM*, 45(8), 61.
- Grover, P., & Kar, A. K. (2017 Sep). Big data analytics: A review on theoretical contributions and tools used in literature. *Global Journal of Flexible Systems Management*, 18(3), 203–229.
- Hai, R., Geisler, S., & Quix, C. (2016). Constance: An intelligent data lake system. In *Proceedings of the 2016 international conference on management of data* (pp. 2097–2100). New York, NY: ACM.
- Han, J., & Kamber, M. (2006). *Data mining: Concepts and techniques*. San Francisco, CA: Morgan Kaufmann.
- Hastie, T., Tibshirani, R., & Friedman, J. (2008). *The elements of statistical learning: Data mining, inference, and prediction*. New York, NY: Springer.
- Hornick, M. F., Marcadé, E., & Venkayala, S. (2006). *Java data mining: Strategy, standard, and practice: A practical guide for architecture, design, and implementation*. San Francisco, CA: Morgan Kaufmann Publishers Inc.
- Jain, A. K., Duin, R. P. W., & Mao, J. (2000). Statistical pattern recognition: A review. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(1), 4–36. Piscataway, NJ: IEEE.
- Japkowicz, N., & Stefanowski, J. (2016). A machine learning perspective on big data analysis. In *Big data analysis: New algorithms for a new society* (pp. 1–31). Cham, Switzerland: Springer International Publishing.
- Jia, Y., Shelhamer, E., Donahue, J., Karayev, S., Long, J., Girshick, R., ... Darrell, T. (2014). Caffe: Convolutional architecture for fast feature embedding. *ArXiv preprint arXiv:1408.5093*.
- Kim, K., Gutesa, S., Dimitrijevic, B., Lee, J., Spasovic, L., Mirza, W., & Singh, J. (2017). Performance evaluation of video analytics for traffic incident detection and vehicle counts collection. In C. Liu (Ed.), *Recent advances in intelligent image search and video retrieval* (pp. 213–231). Cham, Switzerland: Springer.
- Kober, J., Bagnell, J. A., & Peters, J. (2013). Reinforcement learning in robotics: A survey. *The International Journal of Robotics Research*, 32(11), 1238–1274.
- Kotthoff, L., Thornton, C., Hoos, H., Hutter, F., & Leyton-Brown, K. (2016). Auto-WEKA 2.0: Automatic model selection and hyperparameter optimization in WEKA. *Journal of Machine Learning Research*, 17, 1–5.
- Krawczyk, B., Minku, L. L., Gama, J., Stefanowski, J., & Woźniak, M. (2017). Ensemble learning for data stream analysis: A survey. *Information Fusion*, 37, 132–156.
- Landset, S., Khoshgoftaar, T. M., Richter, A. N., & Hasanin, T. (2015). A survey of open source tools for machine learning with big data in the Hadoop ecosystem. *Journal of Big Data*, 2(1), 24.
- Laney, D. (2001). *3D data management: Controlling data volume, velocity and variety*. Stamford, CT: META Group Research Note.
- LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.
- Levine, S., Pastor, P., Krizhevsky, A., & Quillen, D. (2017). Learning hand-eye coordination for robotic grasping with large-scale data collection. In D. Kulić, Y. Nakamura, O. Khatib, & G. Venture (Eds.), *2016 International symposium on experimental robotics* (pp. 173–184). Cham, Switzerland: Springer International Publishing.
- Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., ... Sánchez, C. I. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42, 60–88.
- Liu, H., Chen, S., & Kubota, N. (2013). Intelligent video systems and analytics: A survey. *IEEE Transactions on Industrial Informatics*, 9(3), 1222–1233.
- Liu, L., Hao, T., Xie, Z., Horsman, G. P., & Chen, Y. (2016). Genome mining unveils widespread natural product biosynthetic capacity in human Oral microbe streptococcus Mutans. *Scientific Reports*, 6, 37479.
- Lovell, M. C. (1983). Data mining. *The Review of Economics and Statistics*, 65, 1–11.
- Mikut, R. (2010). Computer-based analysis, visualization, and interpretation of antimicrobial peptide activities. *Methods in Molecular Biology*, 618, 287–299.
- Mikut, R., & Reischl, M. (2011). Data mining tools. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 1(5), 431–443.
- Mikut, R., Reischl, M., Burmeister, O., & Loose, T. (2006). Data mining in medical time series. *Biomedizinische Technik*, 51(5/6), 288–293.
- O’Leary, D. E. (2014). Embedding AI and crowdsourcing in the big data lake. *IEEE Intelligent Systems*, 29(5), 70–73.

- Pandey, R. V., Pabinger, S., Kriegner, A., & Weinhäusel, A. (2017). DaMold: A data-mining platform for variant annotation and visualization in molecular diagnostics research. *Human Mutation*, 38(7), 778–787.
- Pechter, R. (2009). What's PMML and what's new in PMML 4.0? *ACM SIGKDD Explorations Newsletter*, 11(1), 19–25.
- Ravi, K., & Ravi, V. (2015). A survey on opinion mining and sentiment analysis: Tasks, approaches and applications. *Knowledge-Based Systems*, 89, 14–46.
- Reinsel, D., Gantz, J., & Rydning, J. (2017). *Data age 2025: The evolution of data to life-critical*. Framingham, MA: International Data Corporation.
- Rexer, K., Gearan, P., & Allen, H. (2016). *2015 Data science survey*. Winchester, MA: Rexer Analytics.
- Riel, A., & Flatscher, M. (2017). A design process approach to strategic production planning for industry 4.0. In J. Stofa, S. Stofa, R. V. O'Connor, & R. Messnarz (Eds.), *Systems, software and services process improvement: 24th European conference, EuroSPI 2017, Ostrava, Czech Republic* (pp. 323–333). Cham, Switzerland: Springer International Publishing.
- Rodríguez-Mazahua, L., Rodríguez-Enríquez, C.-A., Sánchez-Cervantes, J. L., Cervantes, J., García-Alcaraz, J. L., & Alor-Hernández, G. (2016). A general perspective of big data: Applications, tools, challenges and trends. *The Journal of Supercomputing*, 72(8), 3073–3113.
- Runkler, T. A. (2016). *Data analytics: Models and algorithms for intelligent data analysis*. Springer Vieweg: Wiesbaden, Germany.
- Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., ... Fei-Fei, L. (2015). Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3), 211–252.
- Sallam, R. L., Howson, C., Idoine, C. J., Oestreich, T. W., Richardson, J. L., & Tapadinhas, J. (2017). *Magic quadrant for business intelligence and analytics platforms*. Stamford, CT: Gartner RAS Core Research Notes Gartner.
- Sapountzi, A., & Psannis, K. E. (2016). *Social networking data analysis tools & challenges*. Amsterdam, Netherlands: Elsevier.
- Schindelin, J., Rueden, C. T., Hiner, M. C., & Eliceiri, K. W. (2015). The ImageJ ecosystem: An open platform for biomedical image analysis. *Molecular Reproduction and Development*, 82(7–8), 518–529.
- Scott, J. (2017). *Social network analysis*. Thousand Oaks, CA: SAGE.
- Seide, F., & Agarwal, A. (2016). CNTK: Microsoft's open-source deep-learning toolkit. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining* (pp. 2135–2135). New York, NY: ACM.
- Shearer, C. (2000). The CRISP-DM model: The new blueprint for data mining. *Journal of Data Warehousing*, 5(4), 13–22.
- Simoff, S., Böhlen, M. H., & Mazeika, A. (2008). *Visual data mining: Theory, techniques and tools for visual analytics*. Berlin, Germany: Springer.
- Smyth, P. (2000). Data mining: Data analysis on a grand scale? *Statistical Methods in Medical Research*, 9(4), 309–327.
- Sonnenburg, S., Braun, M. L., Ong, C. S., Bengio, S., Bottou, L., Holmes, G., ... Williamson, R. (2007). The need for open source software in machine learning. *Journal of Machine Learning Research*, 8, 2443–2466.
- Stahl, F., Gabrys, B., Gaber, M. M., & Berendsen, M. (2013). An overview of interactive visual data mining techniques for knowledge discovery. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 3(4), 239–256.
- Suganuma, M., Shirakawa, S., & Nagao, T. (2017). A genetic programming approach to designing convolutional neural network architectures. In *Proceedings of the genetic and evolutionary computation conference* (pp. 497–504). New York, NY: ACM.
- Sun, G.-D., Wu, Y.-C., Liang, R.-H., & Liu, S.-X. (2013). A survey of visual analytics techniques and applications: State-of-the-art research and future challenges. *Journal of Computer Science and Technology*, 28(5), 852–867.
- Theano Development Team. (2016). Theano: A Python framework for fast computation of mathematical expressions. *ArXiv preprint arXiv: 1605.02688*.
- Vanschoren, J., Braun, M. L., & Ong, C. S. (2014). *Open science in machine learning* *ArXiv preprint arXiv: 1402.6013*.
- Wang, J., Chen, Q., & Yao, J. (2008). Data mining software. In L. A. Tomei (Ed.), *Encyclopedia of information technology curriculum integration* (pp. 173–178). Hershey, PA: IGI Global.
- Wang, J., Hu, X., Hollister, K., & Zhu, D. (2008). A comparison and scenario analysis of leading data mining software. *International Journal of Knowledge Management*, 4(2), 17–34.
- Ward, J. S., & Barker, A. (2013). Undefined by data: A survey of big data definitions. *ArXiv preprint arXiv:1309.5821*.
- Wever, M., van Rooijen, L., & Hamann, H. (2017). Active coevolutionary learning of requirements specifications from examples. In *Proceedings of the genetic and evolutionary computation conference* (pp. 1327–1334). New York, NY: ACM.
- Witten, I. H., Frank, E., Hall, M. A., & Pal, C. J. (2016). *Data mining: Practical machine learning tools and techniques*. Amsterdam, Netherlands: Morgan Kaufmann.
- Wu, X., Zhu, X., Wu, G.-Q., & Ding, W. (2014). Data mining with big data. *IEEE Transactions on Knowledge and Data Engineering*, 26(1), 97–107.
- Wu, Y., Cao, N., Gotz, D., Tan, Y.-P., & Keim, D. A. (2016). A survey on visual analytics of social media data. *IEEE Transactions on Multimedia*, 18(11), 2135–2148.
- Ziemert, N., Alanjary, M., & Weber, T. (2016). The evolution of genome mining in microbes—A review. *Natural Product Reports*, 33(8), 988–1005.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

How to cite this article: Bartschat A, Reischl M, Mikut R. Data mining tools. *WIREs Data Mining Knowl Discov*. 2019;e1309. <https://doi.org/10.1002/widm.1309>